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Prediction and optimization of hardness in AlSi10Mg alloy produced by laser powder bed fusion using statistical and machine learning approaches

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The primary objective of this study is to evaluate the influence of critical process parameters on the hardness of AlSi10Mg alloy fabricated via the Laser Powder Bed Fusion (LPBF) technique. To optimize these parameters, a Taguchi-based signal-to-noise (S/N) ratio analysis was employed. Furthermore, Analysis of Variance (ANOVA) was conducted to quantify the statistical significance and contribution ratio of each parameter to the observed variation in hardness. In addition, Machine Learning algorithms were applied to predict hardness values. A multidisciplinary approach was adopted to optimize production processes, reduce manufacturing costs, and shorten processing times. Taguchi and ANOVA analyses were performed using Minitab 21, while Machine Learning implementations were carried out in R statistical software. The applied Machine Learning techniques include Artificial Neural Networks (ANN), Random Forest (RF), Support Vector Machines (SVM), and Multiple Linear Regression (MLR). The results indicate that scan speed exerts the most significant influence on hardness, followed by laser power. Notably, the highest hardness was achieved using a laser power of 200 W, a hatch distance of 0.20 mm, and a scan speed of 2000 mm/s. Among the models, SVM achieved the highest coefficient of determination (R^2 = 0.73). This study highlights the importance of integrating Machine Learning and statistical analysis methods for the effective modeling and optimization of LPBF processes. The findings contribute significantly to the literature and serve as a valuable reference for future research aimed at improving LPBF process efficiency and performance.

Keywords Laser powder bed fusion, Hardness optimization, Taguchi method, Machine learning in additive manufacturing, Process parameter optimization

Advancing manufacturing technologies enable higher efficiency and customized products in industrial applications. In recent years, Laser Powder Bed Fusion (LPBF) technology has emerged as a significant alternative for the production of metallic parts, offering high precision and the realization of unique designs[1,2]. LPBF is frequently used in the production of aluminum alloys[3,4]. The AlSi10Mg alloy, with its light weight, high durability, and excellent machinability, is widely used, particularly in the automotive and aerospace sectors[5,6]. However, controlling the LPBF process requires consideration of numerous factors that affect material properties[7].

The sensitivity of material properties to manufacturing parameters highlights the need for process optimization[8]. Parameters such as laser power, scan speed, and hatch spacing play a critical role in determining the quality of the final product. In this context, systematically analyzing the effects of manufacturing parameters makes it possible to produce parts with high quality and homogeneous properties[9].

Statistical analysis methods such as Taguchi and ANOVA are commonly used in these optimization processes. The Taguchi method systematically evaluates the effects of parameters, while ANOVA analysis helps determine the statistical significance of factors. The combination of these methods provides a powerful tool for optimizing manufacturing processes. The systematic approach offered by the Taguchi method in identifying the effects of manufacturing parameters and obtaining optimal parameter sets has been widely used in the literature[10–12]. However, further research is needed to fully understand the interactions of the parameters in the LPBF process[13].

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Additive Manufacturing (AM) contributes significantly to making production processes smarter, more efficient, and more predictable. Advanced Machine Learning (ML) algorithms play a crucial role in analyzing the large-scale and multi-dimensional datasets generated during AM processes, offering innovative and effective solutions in various areas such as process parameter optimization, anomaly detection and design validation[14–16]. The integration of these algorithms into AM systems not only enhances productivity and quality but also reduces production costs and facilitates the reliable and rapid implementation of customized design approaches. In particular, deep learning-based models enable high-accuracy decision support systems in quality control by analyzing real-time data collected through in-situ sensors during the manufacturing process, thereby allowing for early prediction of defects and minimizing time and resource waste. Furthermore, by dynamically optimizing process parameters, these algorithms ensure consistency in product quality while increasing process efficiency and reliability. Aligned with the vision of Industry 4.0, ML based applications have become strategic tools that support the sustainability and global competitiveness of additive manufacturing technologies[17].

Dejene and colleagues have examined the key process parameters influencing the surface roughness of AlSi10Mg parts produced via the LPBF method. They compared the performance of various ML models, specifically Random Forest Regression (RFR) and Support Vector Regression (SVR), in predicting the Ra and Rq values. The findings indicate that layer thickness has the most significant effect on surface roughness. Additionally, the RFR model outperformed the SVR model in terms of accuracy for both surface roughness metrics. This study highlights the critical role of process parameter optimization and the selection of appropriate prediction models in improving surface quality[18]. Dejene and Lemu investigated the influence of process parameters on the mechanical properties of AlSi10Mg parts fabricated via LPBF and evaluated the predictability of these properties using ML models. The findings highlight the importance of parameter optimization in mitigating anisotropy and demonstrate the effectiveness of RFR and SVR models in achieving high prediction accuracy with limited datasets[19].

In this study, four different ML algorithms were applied to predict the hardness values obtained from the LPBF process: Artificial Neural Networks (ANN), Random Forest (RF), Multiple Linear Regression (MLR), and Support Vector Machine (SVM). As noted by Qi et al., considering the significant potential of ML in optimizing process parameters and microstructure design in LPBF, this approach allows for more accurate predictions by considering the multiple effects of manufacturing parameters[13]. The continuous advancements in ML and computational technologies are expected to help us better understand the dynamic processes of LPBF and other metal-based additive manufacturing techniques. As emphasized by Wang et al., the progress of these technologies presents a significant opportunity for a better understanding and control of the processes[20].

In this context, Shehbaz and Peng also highlight the important role of ML techniques in additive manufacturing processes. They state that specialized ML algorithms have great potential in additive manufacturing in terms of innovation and efficiency, and emphasize the critical role of collaboration between academia and industry in the development of MLbased approaches[21].

The ML techniques used in our study, considering the diversification of datasets and the creation of reliable data sources, contribute significantly to the optimization of mechanical properties in the LPBF process. In the work of Hodroj et al., ML techniques were employed to predict density using data obtained from the literature, and significant findings regarding the accuracy of these predictions were reported[22]. Similarly, Agnihotri et al. utilized experimental data from the relevant literature to train and test their model. The process parameters were optimized to achieve the desired part density[23]. However, by generating our own data and contributing to this literature, a larger dataset can be obtained, allowing for more accurate and effective modeling of LPBF processes. In this context, ML approaches, as an alternative to traditional engineering analyses, hold great potential for more efficient and predictable management of manufacturing processes.

The main objective of this study is to determine the effects of process parameters on the hardness of the AlSi10Mg alloy during LPBF, identify optimal parameter sets using Taguchi method, and evaluate the significance of these parameters through ANOVA analysis. Additionally, hardness predictions are made using ML algorithms. This versatile approach will contribute significantly to optimizing manufacturing processes, reducing costs, and shortening process times. The structure of the study is organized as follows: In the Materials and Methods section, the chemical composition of the AlSi10Mg powder and the production processes are discussed in detail. A full factorial experimental design was employed, with parameter combinations generated through experimental runs, and the approach for analyzing the results obtained from these combinations is explained. In the Results and Discussion section, the obtained microhardness values and the effects of process parameters on these values are presented, and these findings are compared with other studies in the literature. Additionally, the accuracy of ML algorithms in hardness prediction is examined, and the prediction accuracy of these methods is evaluated. The Conclusion section provides a summary of the main findings of the study, highlighting the contributions made to hardness optimization in LPBF processes. Suggestions for future research are also provided, emphasizing the need for expanding the dataset and conducting more comprehensive analyses, along with recommendations for new areas of study.

Materials and methods

The AlSi10Mg powder used in this study was sourced from ADDVALUE Company. The chemical composition of the powder can be referenced in[8,24]. The production processes were carried out using the Concept Laser-M2 CUSING 3D printer located at the Additive Manufacturing Technology Application and Research Center at Gazi University. The experiments were planned based on a full factorial experimental design, incorporating the parameter combinations outlined in Table 1. These parameters were determined by considering the relevant literature[25–27] and expert opinions. A constant layer thickness of 0.04 mm was selected, with a scan rotation angle of 67° applied between each layer. The laser diameter was set to 0.10 mm, and the preheating temperature was maintained at 120 °C. The build orientation was defined as 45°, indicating that the part was positioned at

Factors	Unit	Variable levels		
Laser Power	W	200	275	350
Hatch Distance	mm	0.08	0.14	0.20
Scan Speed	mm/s	800	1400	2000

Table 1. Process parameters for the experimental design.

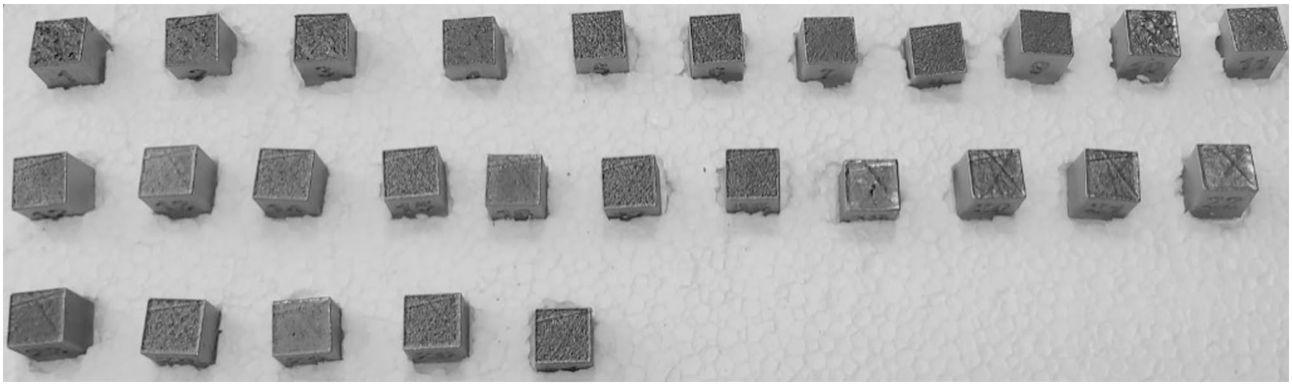

Fig. 1. Appearance of the fabricated parts.

a 45° angle relative to the horizontal plane (X–Y plane) during the fabrication process.The scanning style used was continuous. The produced samples had dimensions of $10\times10\times10$ mm, with a support height of 5 mm. The appearance of the fabricated parts is shown in Fig. 1.

The microhardness measurements were conducted using the EMCO TEST DUROSCAN model device, located at the Technology Research and Application Center of Erciyes University. The Vickers microhardness test was performed on 27 samples, each subjected to a 100 g load for 15 s. Each sample was examined three times to calculate the average hardness value.

Based on the experimental results, Taguchi signal-to-noise (S/N) analysis conducted using Minitab (version 21.2) software was applied to optimize the parameters. The parameter levels were determined based on the 'larger is better' approach. ANOVA was then used to examine the independent effects and interactions of the parameters on microhardness.

R (version 4.4.3) statistical software was used to perform hardness predictions using various ML techniques. Four different methods were employed to predict the hardness of the fabricated samples: ANN, SVM, RF and MLR. The objective is to select the model that best captures the complex relationships between the parameters and material hardness.

To evaluate the generalization capability of the models, the dataset was split into 80% training and 20% testing subsets. The training set was used to enable the models to learn the relationships between the input parameters and the target variable (Microhardness), while the test set was employed to assess predictive performance on previously unseen data.

In the SVM model, a type of kernel known as the Radial Basis Function (RBF) kernel has been utilized. This kernel allows for the separation of data that cannot be linearly divided by mapping it into a higher-dimensional space where better separation can be achieved. Additionally, the C parameter is set to 1.0. This parameter determines the tolerance for errors made during training and helps to prevent overfitting. To optimize the model's performance, hyperparameter tuning was performed using Bayesian optimization. Bayesian optimization was preferred over traditional methods like grid search and random search due to its ability to intelligently select promising parameter combinations based on prior evaluations, making it a more suitable and accurate approach.

In the Random Forest model, the number of trees was set to 100, a commonly used value that typically provides good performance. While increasing the number of trees can enhance the model's accuracy and stability, it may also lead to longer training times and, in some cases, overfitting. Thus, although 100 trees is a reasonable starting point, further optimization through cross-validation could help determine the ideal number of trees for this specific dataset, balancing model performance and computational efficiency.

In the ANN model, a two-hidden-layer structure has been chosen for data learning. The first hidden layer contains 5 neurons, while the second hidden layer consists of 3 neurons. These numbers of layers and neurons have been determined to optimize the model's accuracy and enhance its learning capacity. Typically, these values are selected through a trial-and-error approach to achieve the best possible outcome for a specific problem. This structure aims to balance the complexity of the model with the efficiency of the learning process.

In the case of Multivariate Linear Regression, the regression equation is formulated as follows:

$$\text{Microhardness} = 95.43 - 0.22 \times \text{Laser Power} - 15.03 \times \text{Hatch Distance} + 0.34 \times \text{Scan Speed} \quad (1)$$

These coefficients represent the relationship between the dependent variable (Microhardness) and the independent variables (Laser Power, Scan Distance, and Scan Speed). Each coefficient reflects the magnitude and direction of the effect of the corresponding factor on the target variable.

Results and discussion

The average microhardness values for each specimen are presented in Table 2. The highest microhardness value of 126.00 HV was obtained in experiment 9, which involved with a laser power of 200 W, a hatch distance of 0.20 mm, and a scan speed of 2000 mm/s. Conversely, the lowest microhardness value, 83.37 HV, was observed for specimen 19, which was produced with a laser power of 350 W, a hatch distance of 0.08 mm, and a scan speed of 800 mm/s.

The effect of the process parameters on the microhardness of the specimens has been investigated. A graphical representation illustrating the impact of the controlling factors affecting microhardness is presented in Fig. 2. As the laser power increases, the hardness of the material decreases. Similarly, higher scan speeds result in a reduction in hardness, while beyond a certain threshold, further increases in scan speed lead to an enhancement in hardness. A similar trend is observed with hatch distance. Solic et al. reported that an increase in laser power resulted in a decrease in the microhardness of the produced samples[5]. Chen et al. demonstrated that scanning speed influences microhardness, with higher scanning speeds leading to increased microhardness[28]. Upon examining Fig. 2; Table 3, it can be observed that the highest microhardness value is achieved under the process parameters of 200 W laser power, 0.20 mm hatch distance, and 2000 mm/s scan speed.

ANOVA was applied to determine the contribution rates of the control factors on microhardness. The ANOVA results for the microhardness experiments are presented in Table 4. The contribution percentages of the factors-laser power, hatch distance, and scan speed-on microhardness were found to be 33.2%, 15.6%, and 49.3%, respectively. Furthermore, the ranking in the S/N table provided in Table 3 indicates that scan speed has the highest influence. In the study conducted by Praneeth et al., laser power and scan speed were identified as the most influential parameters affecting hardness[25]. Similarly, Alamri et al. have demonstrated through feature importance analysis using artificial neural networks that laser power and scan speed are the most significant factors affecting hardness[29].

The results of the ML methods for predicting material hardness are presented in Fig. 3, compared with the actual values. The performance of the models is based on their ability to predict the relationship between the experiment numbers (X-axis) and hardness values (Y-axis). In the graphs, the black line represents the actual hardness values, while each colored line (blue, green, red, purple) represents the predictions of the corresponding

Exp. no	Laser power	Hatch distance	Scan speed	Microhardness
1	200	0.08	800	125.33
2	200	0.08	1400	115.33
3	200	0.08	2000	115.67
4	200	0.14	800	116.33
5	200	0.14	1400	105.00
6	200	0.14	2000	107.00
7	200	0.20	800	118.67
8	200	0.20	1400	108.33
9	200	0.20	2000	126.00
10	275	0.08	800	104.23
11	275	0.08	1400	109.33
12	275	0.08	2000	107.67
13	275	0.14	800	113.67
14	275	0.14	1400	108.33
15	275	0.14	2000	119.67
16	275	0.20	800	113.33
17	275	0.20	1400	115.93
18	275	0.20	2000	112.00
19	350	0.08	800	83.37
20	350	0.08	1400	101.10
21	350	0.08	2000	114.67
22	350	0.14	800	93.40
23	350	0.14	1400	110.00
24	350	0.14	2000	123.33
25	350	0.20	800	90.57
26	350	0.20	1400	102.67
27	350	0.20	2000	99.13

Table 2. Average microhardness values and process parameters for each specimen.

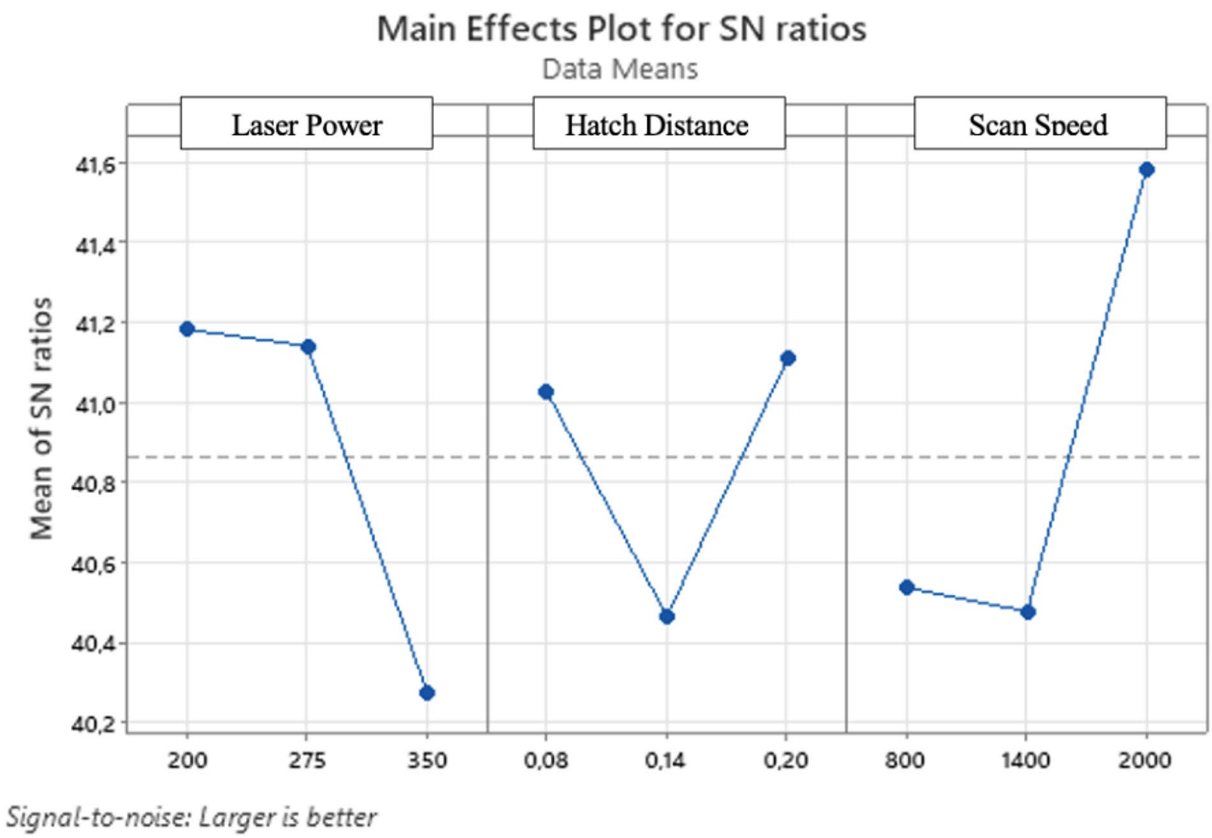

Fig. 2. Main effects plots for microhardness.

Level	Laser power	Hatch distance	Scan speed
1	**41.18**	41.03	40.54
2	41.14	40.46	40.48
3	40.27	**41.11**	**41.59**
Delta	0.91	0.64	1.11
Rank	2	3	1

Table 3. S/N (Signal-to-Noise) table and effect rankings.

Source	DF	Seq SS	Adj SS	Adj MS	F	*p*
Laser Power	2	1.5731	1.5731	0.7865	17.55	0.050
Hatch Distance	2	0.7379	0.7379	0.3689	8.23	0.108
Scan Speed	2	2.3360	2.3360	1.1680	26.06	0.037
Residual Error	2	0.0896	0.0896	0.0448		
Total	8	4.7366				

Table 4. ANOVA results and effects of control factors. S = 0.2117; R-Sq = %98.11; R-Sq(adj) = %92.43

model. The SVM model provided the most accurate and consistent predictions across the entire dataset. While the ANN and MLR models performed well in certain regions, they failed to capture the accuracy during abrupt changes. The RF model successfully followed the overall trend but struggled with smoothing sharp drops and predicting outliers. This behavior of the RF model may stem from its tree-based structure. Since Random Forest operates by averaging the predictions of multiple decision trees, it naturally tends to smooth out abrupt changes. While this characteristic enhances the overall stability of the model, it may reduce sensitivity to sharp

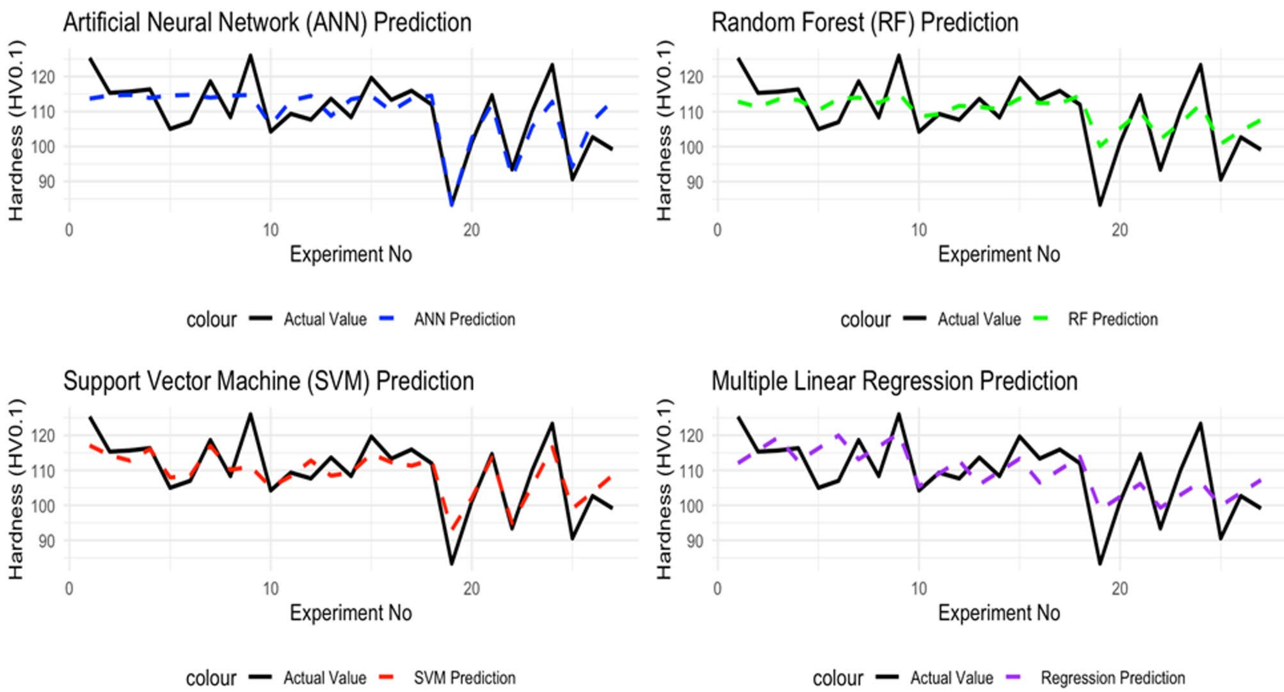

Fig. 3. Material Hardness Predicted by ML Methods and Comparison with Actual Values.

Model	R^2	RMSE
ANN	0.62	6.145304
RF	0.55	6.747351
SVM	0.73	5.255266
MLR	0.38	7.86610

Table 5. Performance evaluation of hardness prediction models.

fluctuations. Overall, the SVM model outperformed the others, particularly in predicting abrupt changes and outlier values.

ML techniques were evaluated in terms of their accuracy and reliability for predicting material hardness, using fundamental metrics such as R^2 (coefficient of determination) and RMSE (root mean squared error). These metrics provide valuable insights into the predictive performance of each model, offering a comprehensive assessment of their effectiveness. The relevant performance metrics are presented in Table 5, which also supports the visual findings illustrated in Fig. 3.

Additionally, the correlation graphs for each model illustrate the relationship between the predicted and actual material hardness and are provided in Fig. 4. Points closer to the diagonal line indicate a higher degree of agreement between the predicted and actual values, reflecting the model's accuracy in predicting material hardness.

The performance of the ML models was evaluated using R^2 and RMSE metrics, as presented in Table 5. The SVM model demonstrated the best performance in predicting material hardness, achieving the highest R^2 value of 0.73 and the lowest RMSE of 5.255266, indicating a strong correlation between predicted and actual hardness values with minimal prediction errors. The ANN model followed with an R^2 of 0.62 and an RMSE of 6.145304, showing moderately good performance. The RF model, with an R^2 of 0.55 and an RMSE of 6.747351, exhibited lower accuracy compared to SVM and ANN, while the MLR model performed the worst, with an R^2 of 0.38 and an RMSE of 7.86610, reflecting significant discrepancies between predicted and actual values. To further validate these results, 5-fold cross-validation was conducted, yielding comparable performance metrics, confirming the models' generalization ability despite the limited dataset size of 27 experiments.

The superior performance of the SVM model can be attributed to its radial basis function (RBF) kernel, which effectively captures the non-linear relationships between process parameters and hardness, particularly the dominant effect of scan speed (49.3% contribution, as per ANOVA results in Table 4). In contrast, the MLR model's poor performance suggests that linear models are less suited for the complex interactions in LPBF processes, as evidenced by the improved R^2 of 0.55 achieved through polynomial regression. In this study, the parameter importance ranking derived from the ANOVA analysis, when employed to guide feature selection in the RF and SVM models, led to a reduction in the RMSE values of the models. Compared to prior studies,

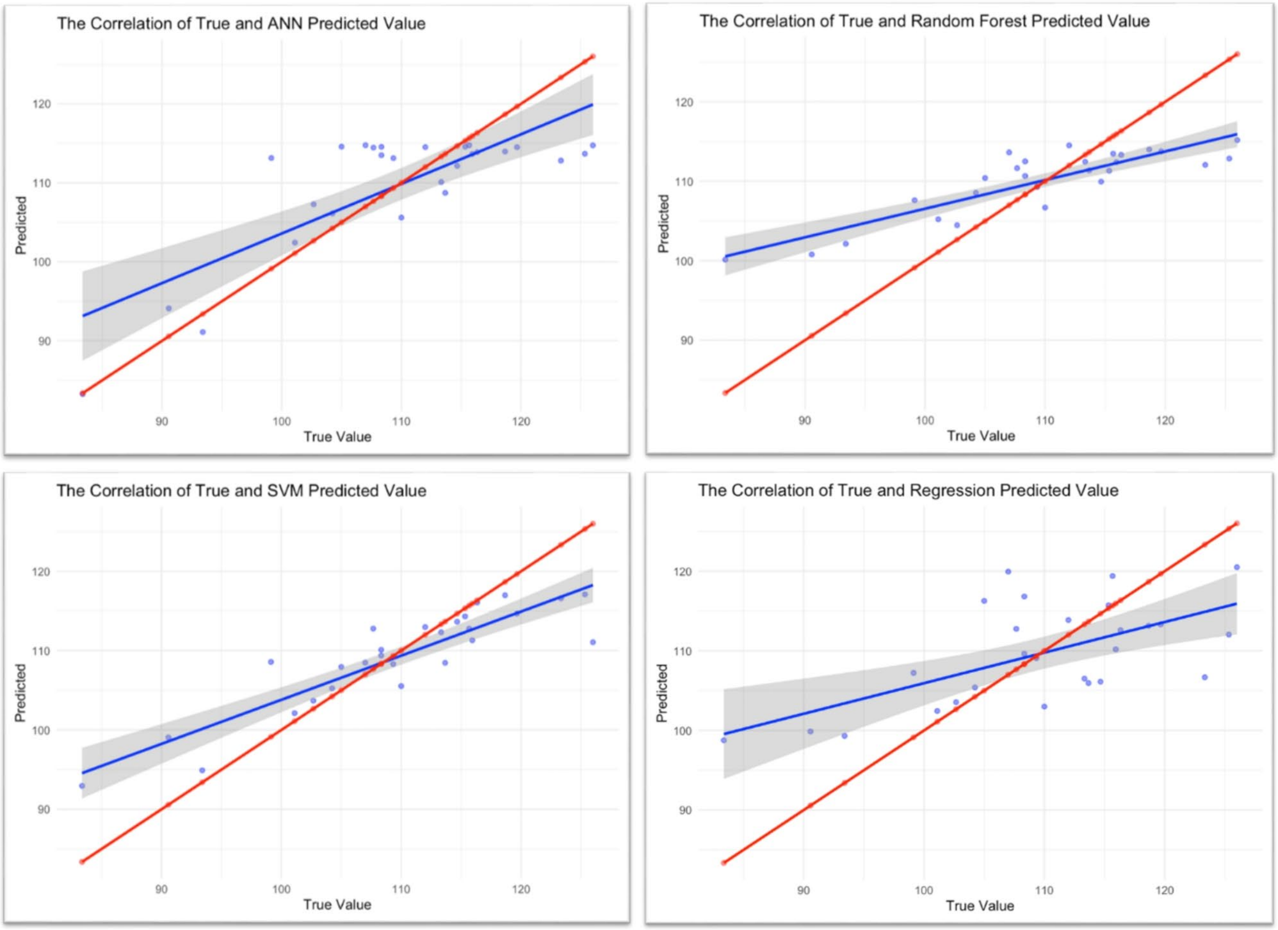

Fig. 4. Correlation graphs of ANN, RF, SVM, and regression models.

such as Hodroj et al.[22], who used SVM for density prediction in LPBF, our findings highlight SVM's versatility in predicting hardness, a critical mechanical property for AlSi10Mg applications. The correlation graphs in Fig. 4 further illustrate these performance differences, with SVM predictions closely aligning with the diagonal line, indicating high accuracy. However, the small dataset size limits the models' generalizability, as noted in the Conclusion. Future studies with larger datasets could enhance model robustness, while integrating real-time sensor data, as suggested by Wang et al.[20], could further improve predictive accuracy in LPBF processes.

Conclusion

This study comprehensively investigated the effects of process parameters on hardness during the production of AlSi10Mg alloy using LPBF and provided valuable insights for process optimization. The results indicate that the scan speed has the greatest influence on hardness, while laser power plays a critical role in determining the final material properties. The highest hardness value was achieved with a combination of 200 W laser power, 0.20 mm hatch distance, and 2000 mm/s scan speed.

The evaluation of ML algorithms for hardness prediction revealed that the SVM model achieved the highest accuracy, with an R^2 value of 0.73 and an RMSE of 5.255266, as shown in Table 5. The ANN model followed with an R^2 of 0.62, while the RF model exhibited a lower R^2 of 0.55, and the MLR model had the lowest performance with an R^2 of 0.38, improved to 0.55 through polynomial regression. The superior performance of SVM can be attributed to its RBF kernel, which effectively captures non-linear relationships, particularly the dominant effect of scan speed (49.3% contribution, Table 4). These findings are supported by 5-fold cross-validation results, which confirm the models' robustness despite the limited dataset. The visualization of these results in Fig. 4, combining predictions from all models, further highlights SVM's accuracy. The use of feature importance scores obtained through ANOVA analysis to guide feature selection in RF and SVM models resulted in a significant improvement in the RMSE values of the models.

The dataset used for training the ML models consisted of only 27 data points, which limits the generalizability of the findings and increases the risk of overfitting, as noted in the ANOVA results (R^2 = 98.11%, Table 4). Therefore, the conclusions should be interpreted within the context of the specific experimental conditions and validated with larger datasets in future studies. Nevertheless, the use of a self-generated dataset underscores the study's contribution to the literature, providing a unique resource for modeling LPBF processes.

Future work could focus on a more comprehensive analysis of the microstructure and mechanical properties of AlSi10Mg alloy produced via LPBF. The effects of different scan strategies, laser modes, and preheating conditions on hardness could be evaluated in detail. Furthermore, the development of multi-objective optimization approaches, considering performance criteria such as surface roughness and toughness in addition to hardness, could lead to more comprehensive production strategies. The integration of real-time data collection through sensor systems during the production process could enable in-situ adjustments, thereby enhancing quality control mechanisms. These advanced studies would contribute to a deeper understanding of the mechanical and microstructural properties of LPBF-produced AlSi10Mg alloy and facilitate more efficient and controlled production processes.

Data availability
All data generated or analyzed during this study are included in this published article.

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Author contributions
The conceptualization, design, data collection, analysis, and writing of this study were solely carried out by İBT.

Declarations

Competing interests
The authors declare no competing interests.

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